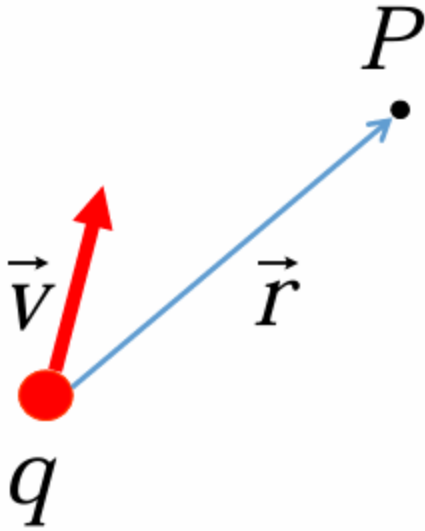


Chapter 29 Magnetic Fields Due to Currents

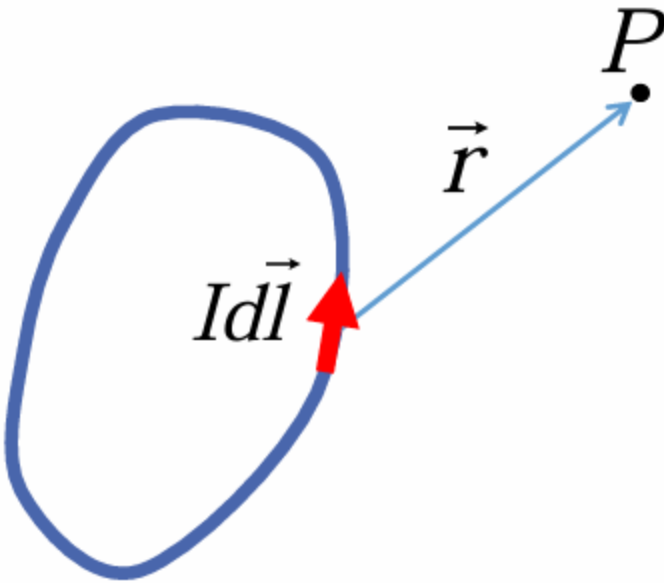
Magnetic Field of a moving charge



$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}$$

where $\mu_0 = 4\pi \times 10^{-7} \text{ N} \cdot \text{s}^2 / \text{C}^2$ (vacuum magnetic permeability), $\hat{r}$ is the unit vector from the charge to field point, r is the distance from the charge to field point.

Magnetic Field of a Current Element



For each point,

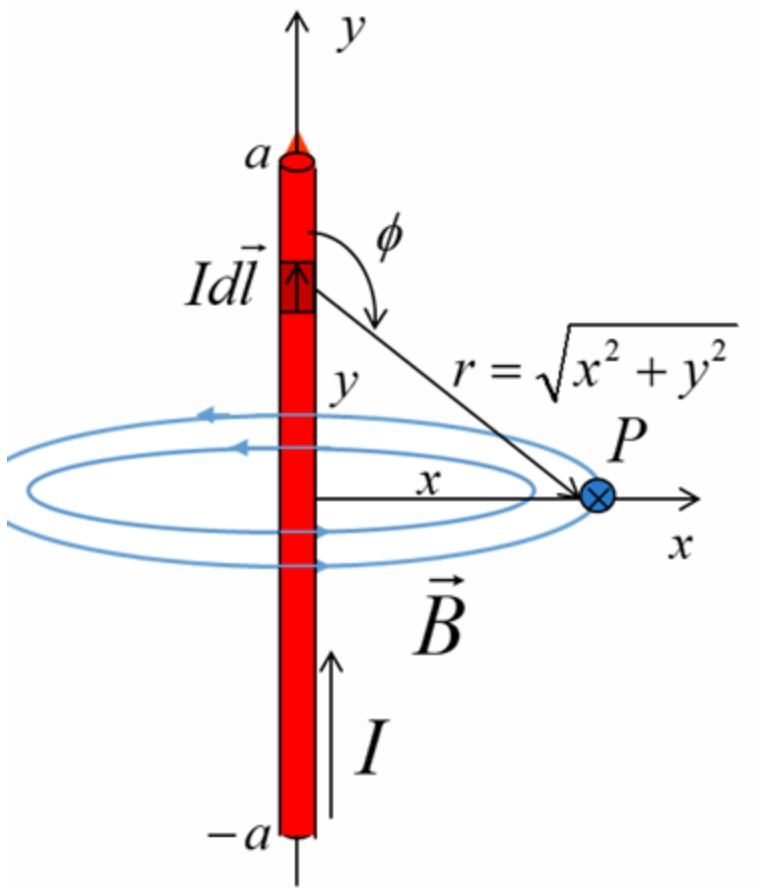
$$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{dq\vec{v} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \cdot \frac{Id\vec{l} \times \hat{r}}{r^2}$$

Thus, the law of Biot and Savart is

$$\vec{B} = \frac{\mu_0}{4\pi} \oint \frac{Id\vec{l} \times \hat{r}}{r^2}$$

In essence, the magnetic field of a current element is the superposition of moving charge.

Magnetic Field due to a current in a Long Straight Wire



It is obtained by the law of Biot and Savart:

$$\vec{B} = \frac{\mu_0}{4\pi} \oint \frac{Id\vec{l} \times \hat{r}}{r^2}, \vec{B} = B\vec{e}$$

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{Idy \cdot \sin \phi}{r^2}, \sin \phi = \frac{x}{r}, r = \sqrt{x^2 + y^2}$$

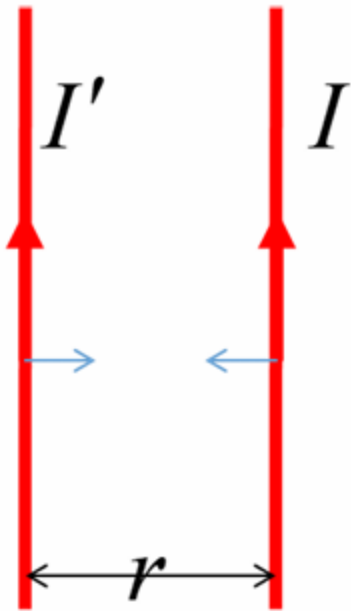
$$B = \oint dB = \frac{\mu_0 I}{4\pi} \int_{-a}^a \frac{xdy}{(x^2 + y^2)^{3/2}} = \frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2 + a^2}}$$

when $a \rightarrow \infty$,

$$\vec{B} = \vec{e} \frac{\mu_0 I}{2\pi x}$$

where $\vec{e}$ is the tangent unit vector of the circle.

Force Between Parallel Conductors



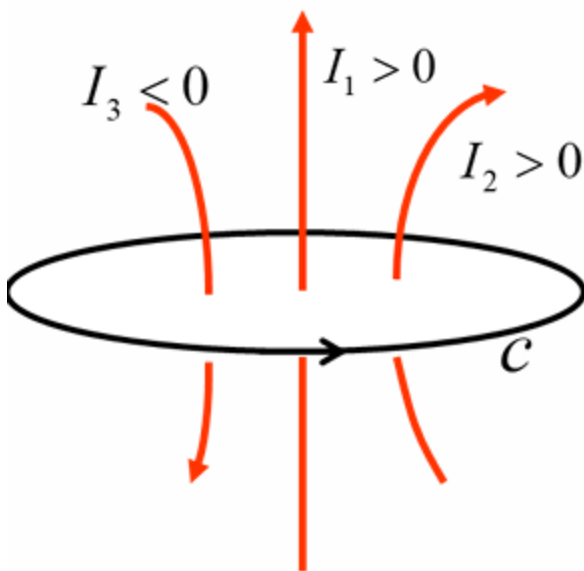
Two straight currents in the same direction attract each other, while those in opposite directions repel each other.

The force per unit length is

$$\frac{F}{L} = \frac{\mu_0 I I'}{2\pi r}$$

Not only do current-carrying conductors in the same direction attract each other, but different parts of the current also attract each other. This constriction of conductive liquids and gases is called pinch effect.

Ampere's Law



$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$$

where i_{enc} is the net current enclosed by the close path.

if the direction of the current follows the **right-hand rule** then the current in a loop is **positive**.

Apply for symmetry situation.